OTA-C 50/60 Hz Notch Filter in Sky130

Current-cancellation OTA style: design from scratch, sizing, verification, power, timing, OTA layout and post-layout analysis

What this document is. A from-scratch account of how this design was sized and checked, using only open-source tools (Xschem, ngspice, Sky130A models). Every number below comes from a simulation in this project or a calculation shown here.

1. Result in one page

Quantity	Result	Target	Status
Notch frequency (tuned bias)	49.96 Hz at Ib = 0.2096 nA; 60.00 Hz at Ib = 0.2535 nA	50 / 60 Hz within 1 Hz	Met
Notch depth (vs 10 Hz gain)	43.1 dB at 50 Hz; 41.6 dB at 60 Hz	≥ 20 dB	Met
Low-frequency gain	−7.06 dB (filter −6.02 dB ideal, plus buffer)	about −6 to −7 dB	Met
Gain above the notch	−12.7 dB at 100 Hz, −14.1 dB at 150 Hz	Not specified	**Limitation**
Stability (2 mV step)	Settles in 43.8 ms, 16.8% overshoot, no growth	No oscillation	Met
Mains rejection in time	50 Hz is 42.3 dB below 10 Hz at the output	≥ 20 dB	Met
THD at 5 mV, 10 Hz	0.18% at the output (0.32% before the buffer)	≤ 2%	Met
Mismatch Monte Carlo (100 runs)	93 runs usable: f_0 49.37 ± 1.78 Hz, depth 42.2 ± 1.4 dB; 7 runs failed	Reported	**93% yield**
Process corners and temperature	f_0 45.3 to 54.2 Hz at fixed bias; retrimmed to 49.7–50.5 Hz by Ib	Reported	Tunable
Power	1.56 µW total: filter 2.32 nW, buffer 1.565 µW	Reported	Buffer dominates
OTA layout	134.4 × 29.9 µm; DRC 0 errors; LVS: circuits match uniquely	Clean	Met
Post-layout (extracted OTA)	49.95 Hz, 41.7 dB at Ib = 0.2124 nA; 60.00 Hz, 40.6 dB; Monte Carlo 94/100 usable	Close to schematic	Met

Honest summary: nominal performance is strong. The two weaknesses are the 7% of mismatch runs in which device offsets push the filter's DC point out of range, and the response above the notch, which stays lower than the low-frequency gain. The OTA layout passes DRC and LVS, and the extracted layout behaves almost exactly like the schematic. Sections 8, 11 and 12 give the details.

2. The specification and why a gm-C filter

Mains interference at 50 Hz (India) or 60 Hz has to be removed from biomedical signals, which contain useful content at very low frequencies. A passive or op-amp RC filter would need very large on-chip resistors and capacitors. In a gm-C filter the resistor is replaced by an OTA, whose output current is

$$I_{out} = g_m (V_+ - V_-)$$

An OTA driving a capacitor C forms an integrator with a time constant set by C/g_m, so its characteristic frequency is

$$f = g_m / (2\pi C)$$

For 50 Hz with C = 4 pF this needs $g_m = 2\pi \times 50 \times 4$ pF = **1.257 nA/V**. That is about a thousand times smaller than an ordinary OTA, which is why the transistors must run in weak inversion at sub-nanoamp currents.

Why not the 6th-order filter from two of the papers?

With equal g_m/C the 6th-order canonical filter has the transfer function denominator

$$D(s) = s^6 + \omega\, s^5 + \omega^2\, s^4 + \omega^3\, s^3 + \omega^4\, s^2 + \omega^5\, s + \omega^6$$

This equals $(s^7 - \omega^7)/(s - \omega)$, so its roots are the seventh roots of unity except $s = \omega$. Two of them have a positive real part: the filter is unstable. An AC simulation still draws a tidy notch because AC analysis cannot detect instability, which is why the papers show a working response. Finite OTA gain does not repair it: with A = 27.4 dB the largest pole real part is still $+0.58\,\omega$. So the design uses the **3rd-order elliptic filter** from the 22 nm paper, which is derived from a passive ladder and is stable by construction.

3. The filter mathematics

Fig. 3 of the paper uses five OTAs. Three act as integrators and two as terminating resistors. With node voltages V_1 (the C_1 node), V_x (the C_{L2} node) and V_o (the output), Kirchhoff's current law gives:

$$g_m(V_i - V_x) - g_m V_1 = s\, C_1\, V_1 + s\, C_2\, (V_1 - V_o)$$

$$g_m(V_1 - V_o) = s\, C_{L2}\, V_x$$

$$g_m V_x - g_m V_o = s\, C_3\, V_o + s\, C_2\, (V_o - V_1)$$

Solving these three equations gives the paper's equation (4); a numerical solve of the node equations matches it to rounding error (a difference of 10^{-16}):

$$V_o/V_i = [1 + s^2\, C_2 C_{L2}/g_m^2] \,/\, [2 + s(C_1+C_3+C_{L2})/g_m + s^2 C_{L2}(C_1+2C_2+C_3)/g_m^2 + s^3 C_{L2}(C_1C_2+C_1C_3+C_2C_3)/g_m^3]$$

Three facts follow directly:

• **Notch frequency:** the numerator is zero at $s = j\omega_0$ when $\omega_0 = g_m/\sqrt{(C_2 C_{L2})}$. With $C_2 = C_{L2} = C$ this is $\omega_0 = g_m/C$.

• **DC gain:** at s = 0 the ratio is 1/2, which is −6.02 dB. That loss is built into a doubly terminated filter.

• **Tuning:** ω_0 depends only on g_m, and g_m is set by the bias current, so 50 Hz and 60 Hz share the same capacitors. The 60 Hz bias is 1.209 times the 50 Hz bias (0.2535 / 0.2096), very close to the expected 1.2.

Capacitors

Sky130 MiM capacitors give about 2 fF/µm². I used 45 × 45 µm for C_2 and C_{L2} (Xschem reports 4.084 pF) and 16 × 16 µm for C_1 and C_3 (0.524 pF).

The simulated notch implies an effective capacitance of g_m/ω_0 = 4.89 pF at the measured g_m of 1.534 nA/V, about 0.80 pF above the drawn 4.08 pF. The extra capacitance is the gate capacitance of the large OTA input transistors at the internal nodes.

4. Weak inversion: the transistor model behind the sizing

Below threshold, the drain current of a MOSFET is exponential in its gate voltage:

$$I_D = I_0\, (W/L)\, \exp(V_{GS} / (n\, V_T))$$

Differentiating gives the single most useful design number:

$$g_m / I_D = 1 / (n V_T)$$

where $V_T = kT/q = 25.85$ mV at 27 °C and n is the subthreshold slope factor. The transconductance depends on current, not on W/L, which is why long, large transistors can be used without changing g_m. Rather than assuming numbers, I measured them in ngspice for W/L = 2/2 µm near 1 nA:

Device	V_{GS} at 1 nA	g_m/I_D	n	g_{mb}/g_m
nfet_01v8	0.339 V	25.9 /V	1.49	0.282
nfet_01v8_lvt	0.245 V	30.6 /V	1.27	not measured
pfet_01v8	0.736 V	21.4 /V	1.81	0.285
pfet_01v8_lvt	0.204 V	23.3 /V	1.66	not measured

A standard PFET needs 0.74 V of gate drive, which leaves little headroom in a 1.8 V supply. That is why this design uses low-Vt PFETs.

5. Sizing the current-cancellation OTA

Each input is shared by two PMOS devices of unequal strength: M_1 (strength 1) and M_N (strength N). Their drains are cross-connected, so the output current is the difference between two nearly equal currents:

$$\text{right node: } i_{M1a} + i_{MNb} \quad \text{left node: } i_{MNa} + i_{M1b} \quad I_{out} = \text{right} - \text{left}$$

A differential input V_d changes each branch by $g_{m,branch} \cdot V_d/2$, with M_N carrying N times the current of M_1. Each side's tail share ($I_b/2$) splits N : 1, which gives:

$$g_m = [(N - 1)/(N + 1)] \cdot (I_b/2) \cdot (g_m/I_D)$$

Note that the paper writes this with a device transconductance g_{m1} defined at $I_b/2$. Reading it as a branch current caused a wrong first estimate in this project, and simulation exposed the mistake.

Worked numbers for the final design (N = 3, simulated $I_b = 0.209$ nA, $g_m/I_D = 27.8$ /V from the operating point):

$$g_m = (2/4) \times (0.1045 \text{ nA}) \times 27.8 \text{ /V} = 1.453 \text{ nA/V}, \text{ versus } 1.529 \text{ nA/V from simulation (5\% apart)}$$

The design went through four cycles, each driven by a measured failure:

Cycle	Sizes (W/L µm, N)	Finding	Change made
1	M1/MN 2/8, N=2; mirror 2/20; I_b 1 nA	g_m 4.66 nA/V, too large; A_0 30 dB	Corrected formula, reduced I_b to 0.28 nA, longer channels
2	Same, I_b 0.279 nA	Notch 41.6 dB, but mismatch: V_{os} σ = 9.75 mV, g_m σ = 14.6%; a Monte Carlo run lost the notch	Larger area and N = 3 (less sensitive cancellation)
3	M1/MN 7/20, N=3; mirror 7/20	V_{os} σ 2.84 mV, g_m σ 4.2%, but depth fell to 18.6 dB	Mirror gate capacitance adds phase at 50 Hz; keep input large, shrink mirror
4 (final)	M1/MN 7/20, N=3; mirror 3/8; tail 7/20	Depth 43.1 dB at 50 Hz	Frozen

Why the mirror matters: the NMOS diode has a parasitic pole at about $g_{m,M2}/(2\pi C_{gs})$. A larger gate lowers that pole, and any phase shift inside the OTA at 50 Hz fills in the notch. A long, narrow mirror is good for matching and bad for speed.

Final OTA performance at I_b = 0.209 nA:

g_m	R_{out}	A_0	CMRR	Linear range (±10% g_m)	Nominal V_{os}	Noise (0.05–300 Hz)
1.529 nA/V	49.95 GΩ	37.7 dB	55.9 dB	−23 to +25 mV	7.61 mV	149.9 µV rms

Nominal V_{os} is a real systematic offset. The output node sits at 0 V while the mirror node sits about 0.63 V lower, so the two branches see different drain voltages.

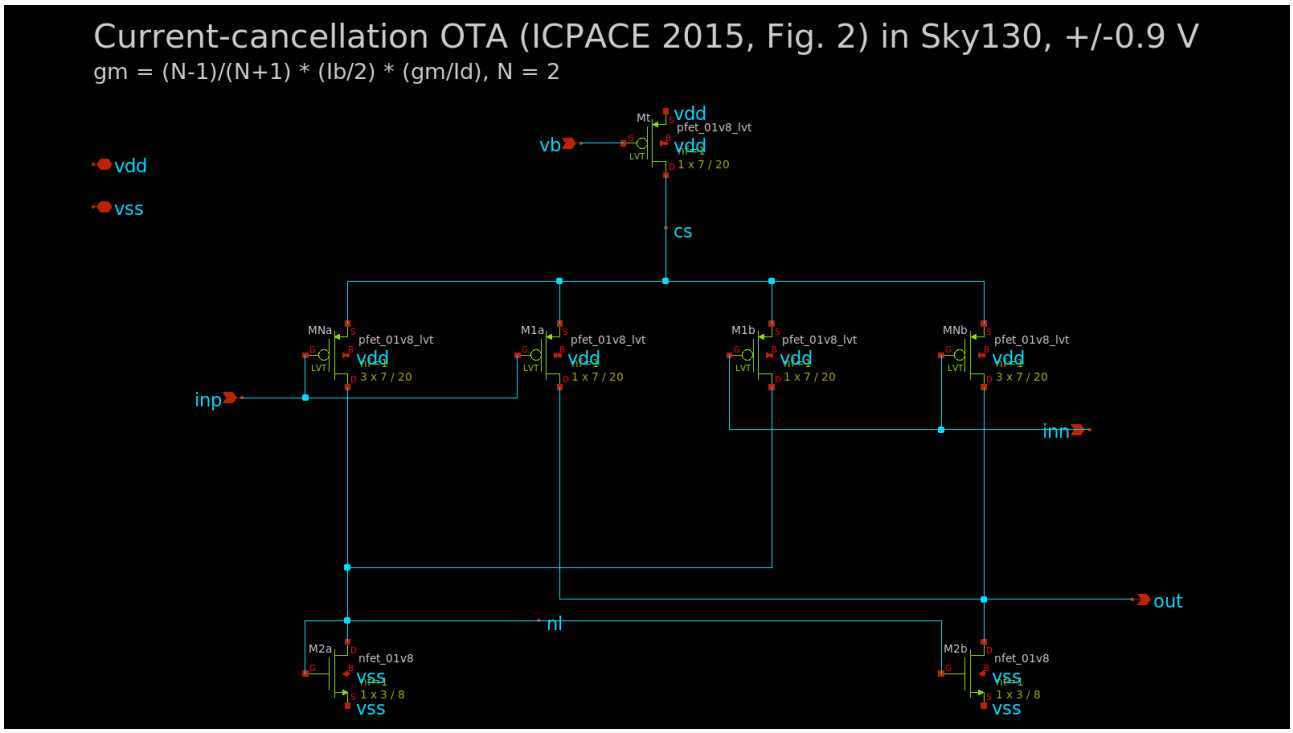


Figure 1. Fully wired current-cancellation OTA in Xschem, with W/L shown on each device.

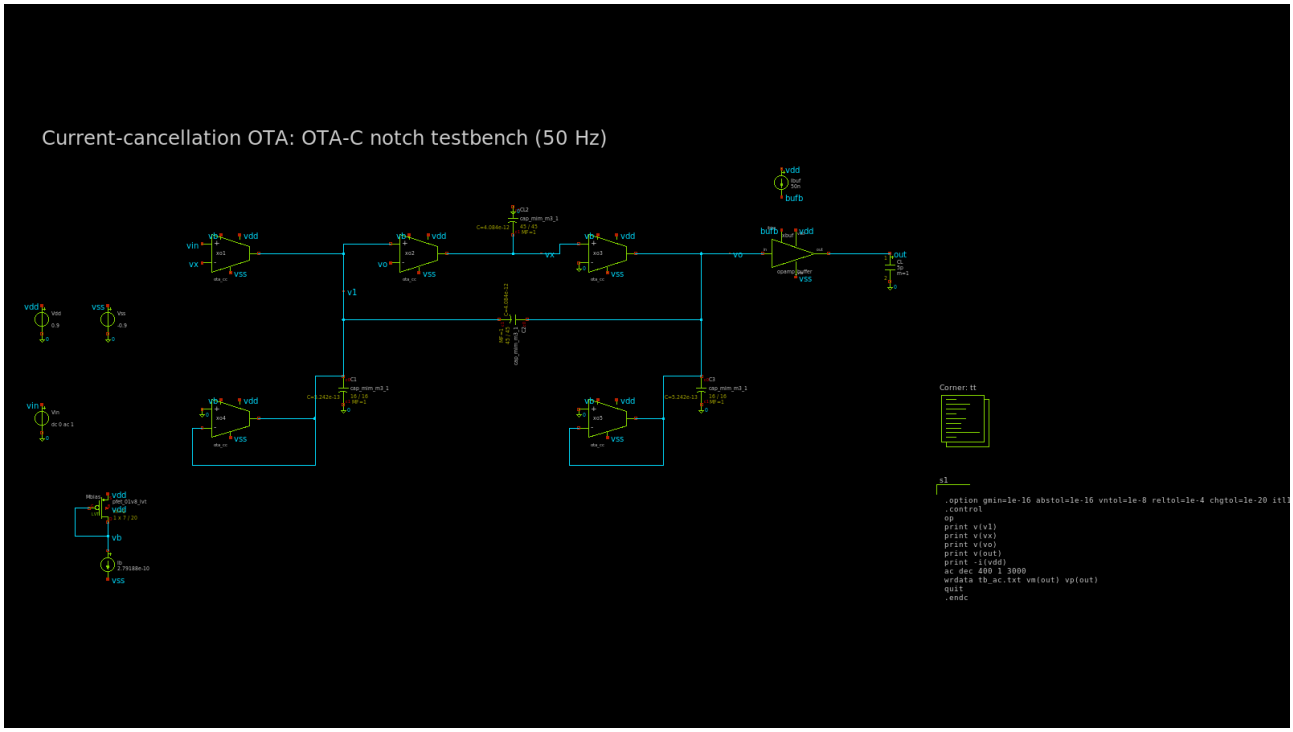


Figure 2. Notch testbench: five OTAs (three integrators, two terminations), Sky130 MiM capacitors, bias diode and the buffer.

6. The op-amp buffer

The paper's output stage has only V_{IN} and V_{OUT}, so it is a unity-gain buffer. It isolates the very high impedance filter output from a load. Its output voltage is set by two balance equations:

$$V_{out} = V_{SG,Mp1} - V_{GS,Mn2} = V_{SG,Mp2} - V_{GS,Mn1}$$

For zero offset those voltage drops must match. In cycle 1 the drops were too large for 1.8 V (headroom failure), so the input devices became low-Vt and wide and the bias dropped to 50 nA. Then the output sat at −134 mV because NMOS body effect raised $V_{GS,Mn2}$ while Mp1 had none. Tying the Mp1 and Mp2 bodies to VDD gives them matching body effect and also removes the need for separate wells. The offset fell to −3.9 mV.

Quantity	Result
Offset	−3.9 mV
DC gain	0.937 V/V (−0.56 dB; body effect in the input followers)
Bandwidth, 5 pF load	23.7 kHz
Step overshoot	0%
Output noise 0.05–300 Hz	198.6 µV rms
Power	1.56 µW

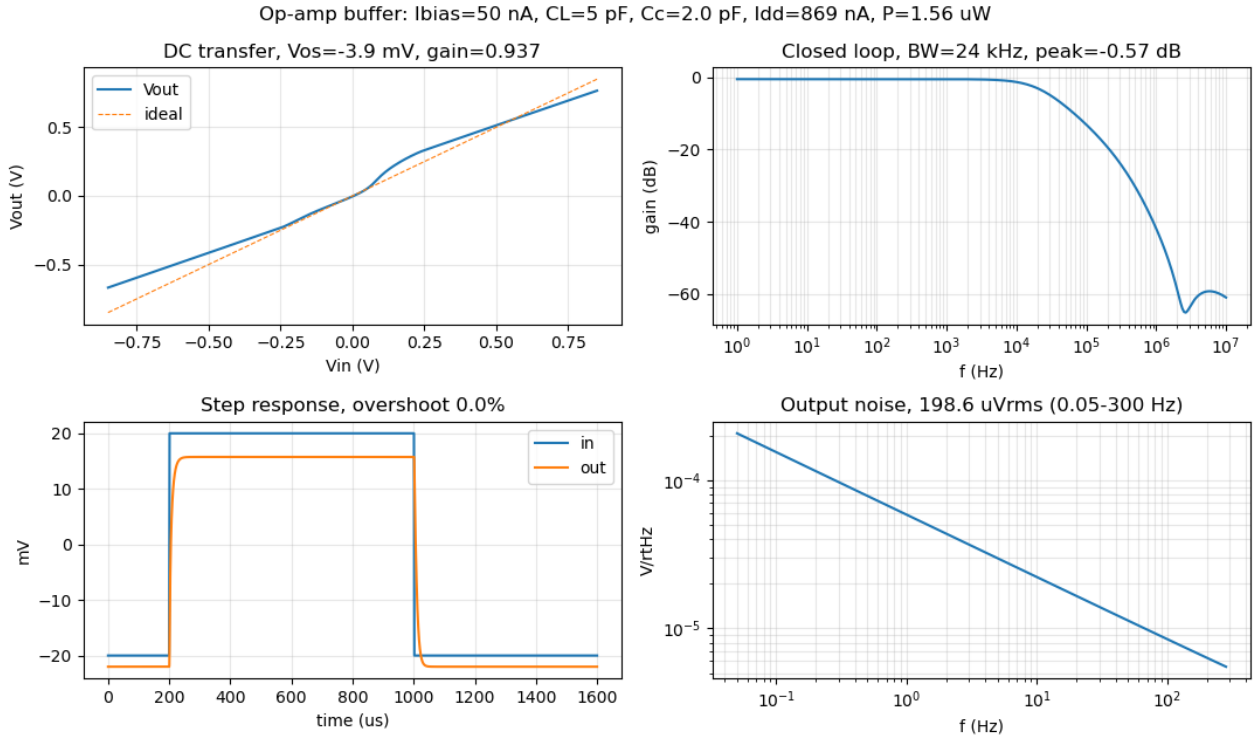


Figure 3. Buffer validation.

7. Simulation method and four traps worth remembering

• **gmin.** ngspice's default of 1 pS is comparable to these OTAs' output conductance (20 pS), so it would distort gain. All runs use gmin = 10^{-16} S with tight current tolerances.

• **m, not mult.** Sky130 multiplies a device through `m=`; `mult=` is silently ignored in a hand-written netlist. Xschem's symbols use the right one.

• **Instance names.** Subcircuit instances must start with x. A name like o1 is read as a transmission line.

• **Phase units.** ngspice's vp( ) returns radians, not degrees.

Every schematic was checked device by device against the design table (terminals, model, W, L, m) with 36 devices and 0 errors, and the netlist Xschem produces is the one that was simulated.

8. Verification results

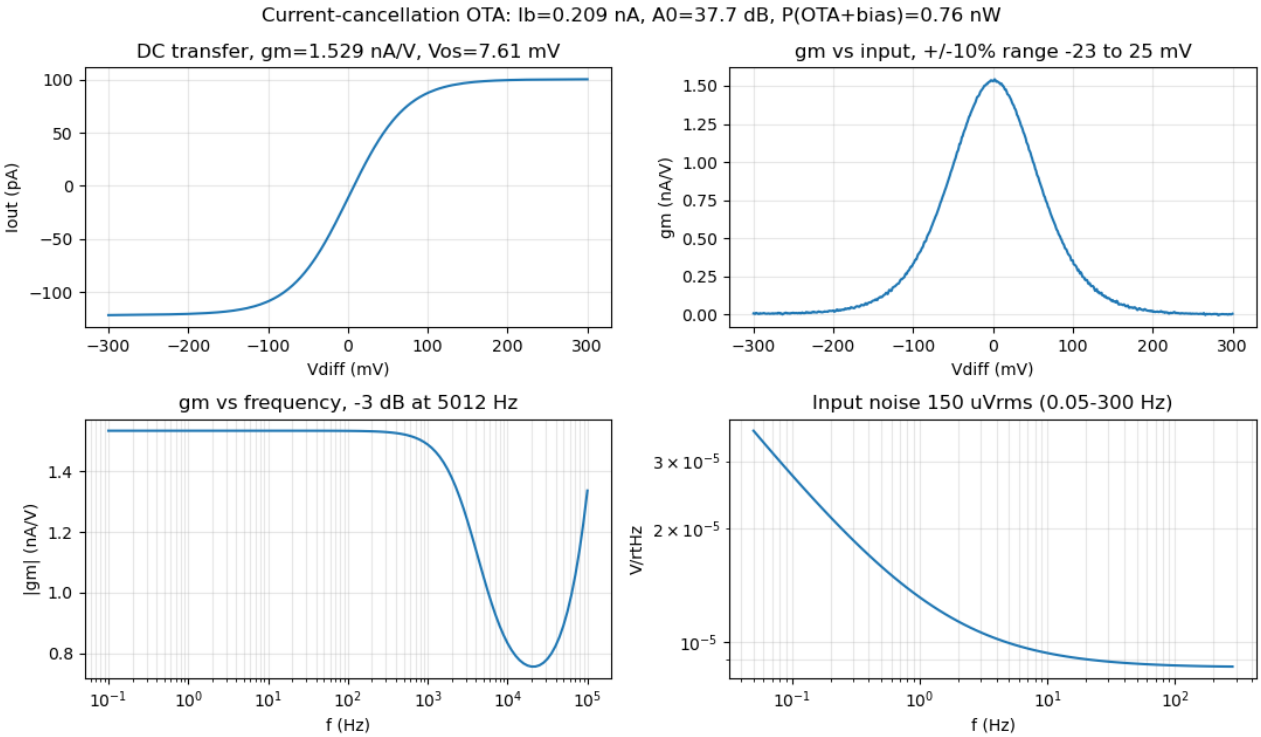


Figure 4. OTA: DC transfer, g_m vs input, g_m vs frequency, and input noise.

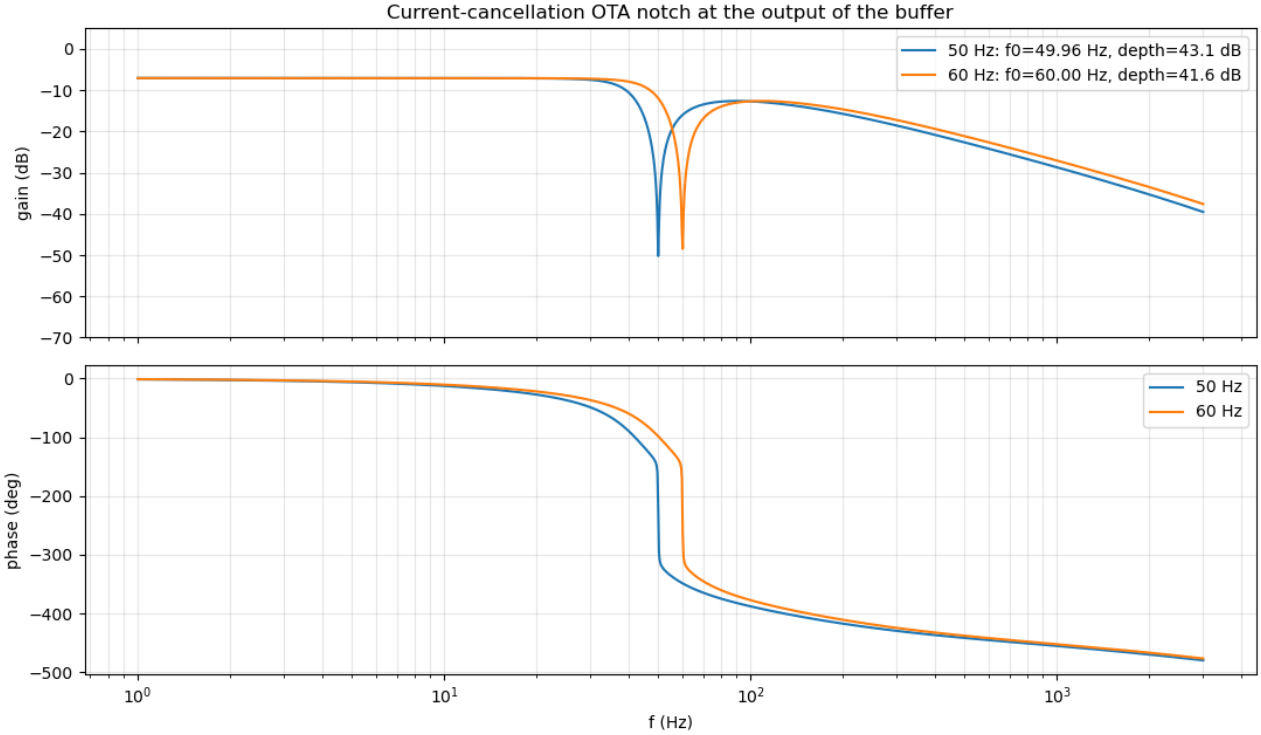


Figure 5. Notch response at the buffer output for 50 Hz and 60 Hz.

Gain above the notch

The filter does not return to its low-frequency gain after the notch:

Frequency	1 Hz	10 Hz	25 Hz	40 Hz	75 Hz	100 Hz	150 Hz	300 Hz
Gain (dB)	−7.05	−7.06	−7.20	−10.55	−13.01	−12.70	−14.09	−18.57

It is a low-pass notch by design, and the OTA parasitics add to C_1 and C_3, lowering the upper pole. An application needing a flat response to 150 Hz would need larger gm and capacitors, or a redesigned upper band.

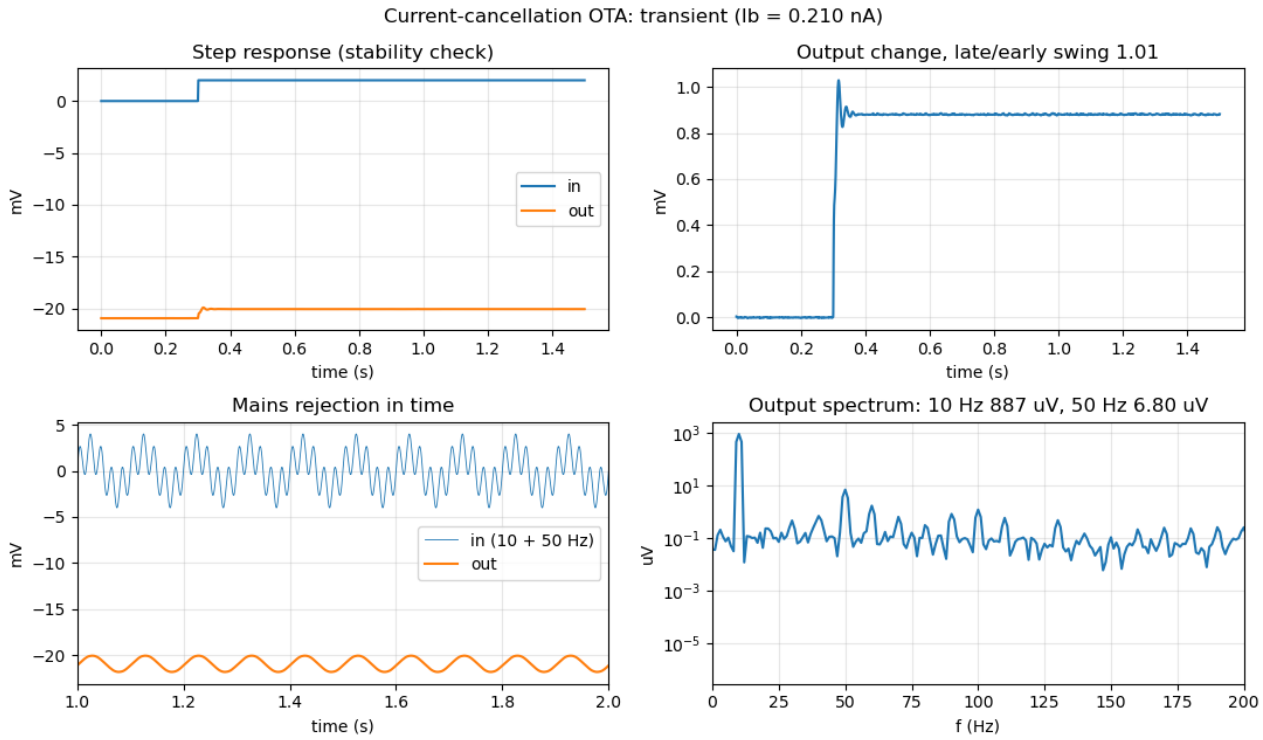


Figure 6. Transient: 2 mV step (stability) and a mix of 2 mV at 10 Hz and 2 mV at 50 Hz (mains rejection).

After the step the output settles within about 0.1 s and then shows only a flat 6–10 µV of solver ripple that does not grow, which confirms stability.

Monte Carlo: device mismatch

100 runs using Sky130's `tt_mm` statistics. The PDK's process Monte Carlo section stops with a model-bin error, so process variation is covered by corners instead. 70% of runs have f_0 within 2 Hz of 50 Hz, 93% within 5 Hz, and 93% have depth of at least 20 dB. Median f_0 is 49.33 Hz.

The raw mean and sigma printed by the simulation script (54.6 and 47.4 Hz) are dominated by the failures, so the table below uses the 93 usable runs. The 7 others are listed separately.

Usable runs	f_0 mean	f_0 σ	Depth mean	Depth σ
93 / 100	49.37 Hz	1.78 Hz	42.2 dB	1.4 dB

Failing seed	f_0 (Hz)	Depth (dB)	Gain 10 Hz (dB)	Output DC (mV)
74	37.3	1.1	-82.9	-634
45	21.8	1.8	-61.8	-584
72	518.9	2.3	-103.0	-670
80	73.6	2.7	-85.7	-649
31	96.3	3.7	-87.7	-647
53	97.6	3.7	-85.1	-652
58	21.0	19.9	-44.4	241

In these runs the output sits around −0.6 V: an offset in one OTA, amplified around the DC feedback, pushes nodes beyond the ±25 mV linear range. Mismatch in this cancellation structure is inherently large because the output is a difference of nearly equal currents.

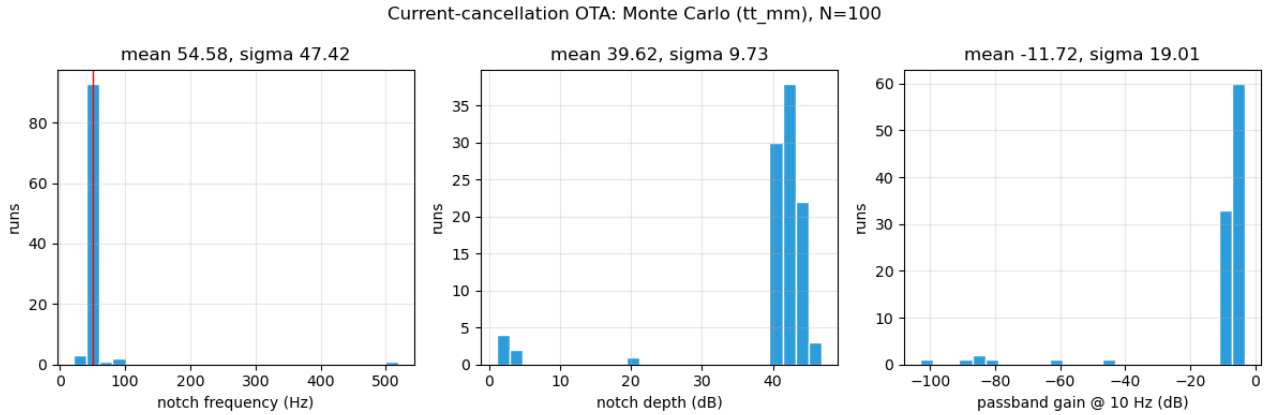


Figure 7. Monte Carlo histograms (including the failing runs).

Process corners and temperature

Corner	T (°C)	f_0 at fixed Ib (Hz)	Depth (dB)	Gain 10 Hz (dB)	Ib trimmed (nA)	f_0 trimmed (Hz)
tt	27	49.96	43.1	−7.06	0.2098	49.98
ss	27	48.54	43.2	−3.05	0.2159	49.95
ff	27	51.76	43.0	−8.18	0.2025	50.17
sf	27	48.04	40.1	−8.27	0.2182	49.96
fs	27	54.15	62.3	−8.66	0.1936	50.49
tt	37	48.83	43.4	−7.71	0.2147	49.96
tt	0	53.23	42.7	+1.89	0.1969	50.19
tt	85	45.31	50.0	−8.24	0.2314	49.67

Bias tuning recovers the notch frequency in every case, which is the practical point of a gm-C design. The passband gain moves a lot at some corners (for example +1.89 dB at 0 °C and −3.05 dB in ss); I did not investigate the reason for this.

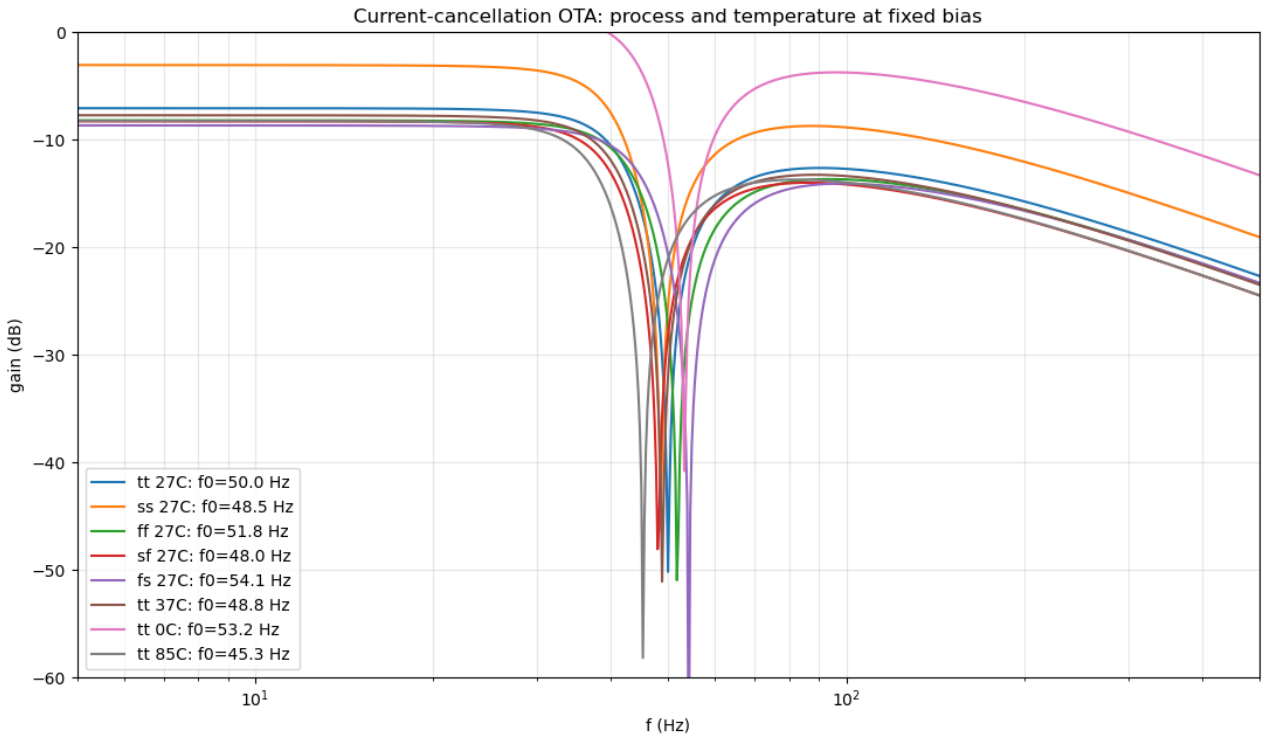


Figure 8. Response across process corners and temperature at fixed bias.

Total harmonic distortion

Measured with a 10 Hz tone, using an FFT over a whole number of cycles after settling, with harmonics 2 to 9:

Input (mV peak)	Output fundamental (mV)	THD at output (%)	THD before buffer (%)	HD2 (%)	HD3 (%)
5	2.218	0.184	0.315	0.145	0.041
10	4.437	0.333	0.632	0.330	0.033
20	8.881	0.658	1.259	0.655	0.058

The 1 mV and 2 mV runs stopped with a simulator timestep error at the start of the sine wave, so those two points are missing. The 5 mV point is the target and is 0.18%, comfortably below the 2% limit. THD is lower at the output than before the buffer, which suggests the buffer's nonlinearity partly offsets the filter's second harmonic.

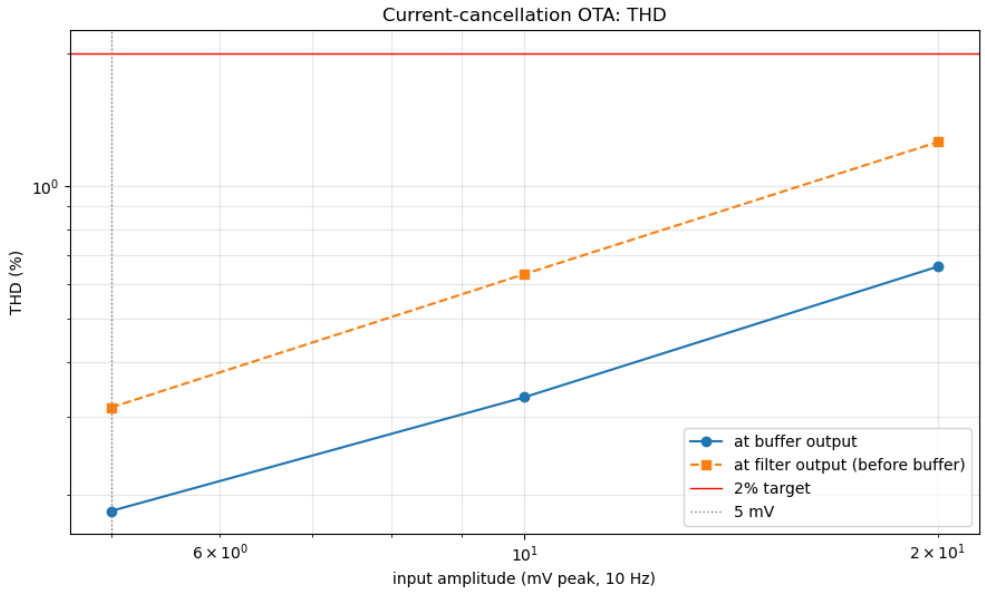


Figure 9. THD vs input amplitude.

9. Power calculation

Static power is simply supply voltage times total current, $P = V_{DD,total} \times I$, with a total supply of 1.8 V (±0.9 V).

Filter (five OTAs plus the bias diode)

$I_{filter} = 5 \times I_{tail} + I_b = 5 \times 0.2156\ nA + 0.2096\ nA = 1.288\ nA$

$P_{filter} = 1.8\ V \times 1.288\ nA =$ **2.32 nW**

The tail carries slightly more than I_b (0.2156 versus 0.2096 nA) because its V_{DS} differs from that of the bias diode.

Buffer, summed over every branch from VDD (operating-point values at I_{bias} = 50 nA)

Branch	Current (nA)
Bias diode Mbp via Mbn2	57.3
Level shifter Mp4	57.9
Level shifter Mn1	50.2
Error-amp tail Mp5	102.0
Load Mp6	32.8
Output stage Mp7	350.0
Sense Mp8	50.6
Stabiliser Mp11	59.4
Stabiliser mirror Mp12	59.0
External Ibias (into Mbn)	50.0
Total	**869.2**

$P_{buffer} = 1.8\ V \times 869.2\ nA =$ **1.565 µW**

Total, as measured

$I_{total} = 866.9\ nA$ from the full testbench; $P_{total} = 1.8\ V \times 866.9\ nA =$ **1.5605 µW**

The filter uses 0.15% of the power. The hand sum for the buffer (869.2 nA) differs slightly from the full circuit because the buffer's input sits at −18.8 mV there and at 0 V in the standalone run. A follow-up can reduce the buffer current substantially, since its 23.7 kHz bandwidth far exceeds the 300 Hz needed.

10. Timing calculation

A filter has no clock, so timing means delay, settling and slew rate.

Quantity	Method	Value
Integrator time constant	$\tau = C_{L2} / g_m = 4.084$ pF / 1.534 nA/V	2.66 ms
Group delay	$\tau_g = -d\varphi/d\omega$, from the AC phase at the output	3.38 ms at 1 Hz; 3.64 ms at 10 Hz; 4.72 ms at 20 Hz; 7.91 ms at 30 Hz
Step rise time, 10–90%	From the 2 mV step in simulation	9.75 ms
Settling to ±2%	From the 2 mV step, measured from the step	43.8 ms
Overshoot	From the 2 mV step	16.8%
OTA maximum output current	$I_{max} = [(N-1)/(N+1)] \times I_{tail} = 0.5 \times 0.2156$ nA	107.8 pA
Slew rate on C_{L2}	$SR = I_{max} / C_{L2}$	26.4 V/s
Slew-limited amplitude at 50 Hz	$V_{max} = SR / (2\pi \times 50$ Hz)	84 mV peak
Buffer bandwidth and step	AC and transient of the buffer	23.7 kHz; within 10% of final in 16.4 µs

Group delay rises toward the notch, as expected: the filter delays frequencies close to 50 Hz more than those far below it. The slew limit of 84 mV peak is well above the 25 mV linear range, so the linear range, rather than slew rate, sets the maximum usable input.

11. OTA layout and post-layout analysis

Only the OTA was laid out; the buffer and the four MiM capacitors stay at schematic level. The OTA is where the design is most sensitive: its output is the difference of two nearly equal currents, so matching between the input devices decides whether the notch survives mismatch.

Layout choices

• **Unit transistors.** Every input device is drawn as an identical PMOS unit of W = 7 µm and L = 20 µm. MN_a and MN_b are each three parallel units and $M1_a$ and $M1_b$ one each, which gives the N = 3 ratio by construction rather than by width.

• **Interleaving.** The eight input units sit in two rows, ordered MN_a MN_b $M1_a$ MN_b over MN_b MN_a $M1_b$ MN_a, so a linear process gradient affects both sides of the cancellation similarly. This is interleaved, not a perfect common centroid: $M1_a$ and $M1_b$ sit in different rows. No dummy devices were added.

• **Routing.** Metal-1 buses (vdd, cs, nl, out) run between the two rows. Source and drain straps on local interconnect drop onto them through mcon contacts. The second row is shifted 0.8 µm so that no strap from one row lines up with a strap from the other. Gates leave on local interconnect to gate buses above and below the rows.

• **Wells and taps.** All PMOS share one n-well tied to VDD by tap rails along the top and bottom. The tail transistor sits at the end of row 1, and the NMOS mirror sits outside the well with its own substrate tap to VSS.

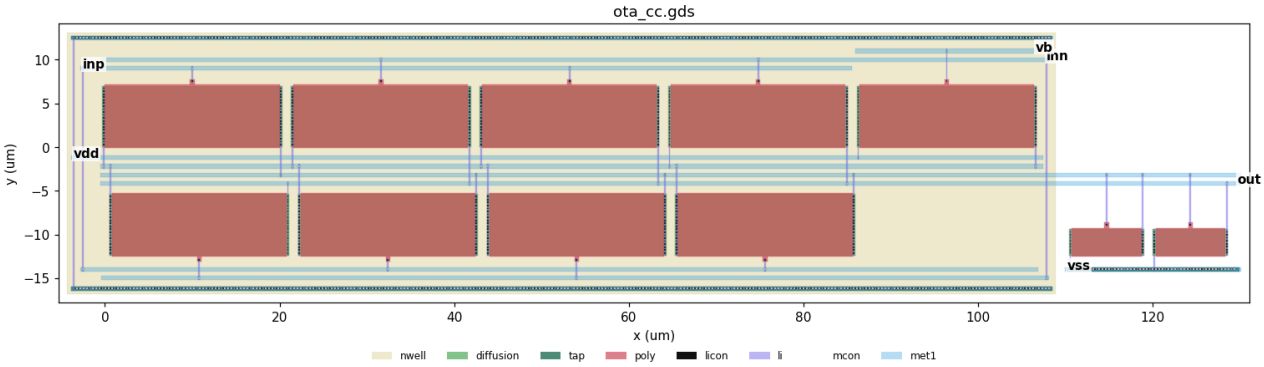


Figure 10. Full OTA layout rendered from ota_cc.gds: two interleaved PMOS rows in the n-well, metal-1 buses between them, NMOS mirror on the right.

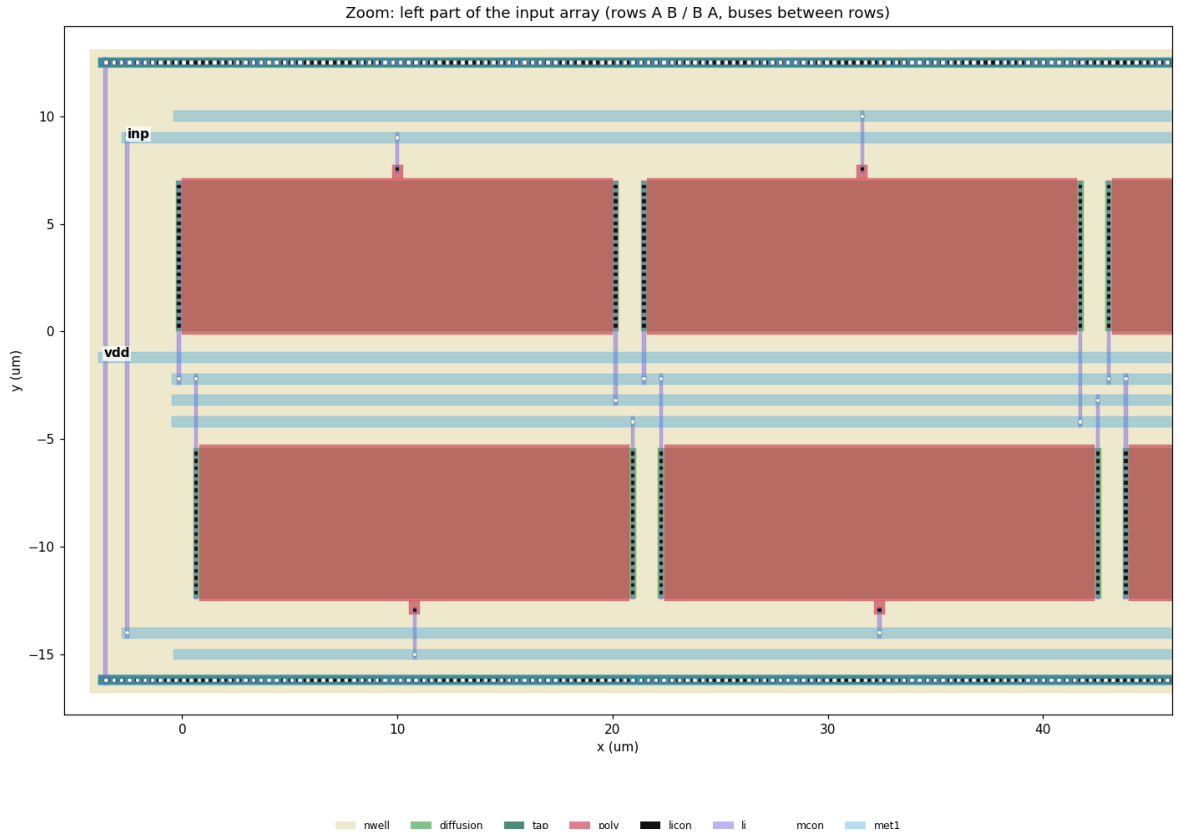


Figure 11. Zoom on the left of the input array showing the unit transistors, source/drain straps, contacts and buses.

Physical verification

The first pass had 22 DRC errors and failed LVS, all in the NMOS mirror. The drain straps there were drawn towards the wrong side, because the mirror's buses sit above it while the PMOS rows' buses sit below. Making every strap cover its contact and reach its bus on either side fixed both. The final result is **DRC: 0 errors** with the full Sky130 rule deck, and **LVS: circuits match uniquely** against the Xschem netlist. Netgen merges the three parallel units of each MN back into m = 3.

Extracted parasitics

Extraction with coupling capacitance gives 555 fF per OTA across 27 capacitors. 419 fF of it is between VDD and VSS, which does nothing to the signal because both are held by supplies. The signal nodes see much less: 29 fF vdd–inn, 23 fF vdd–inp, 13 fF inp–inn and 8.5 fF nl–out. Wire resistance was not extracted; at sub-nanoamp currents it is negligible.

Pre- versus post-layout

Quantity	Schematic	Extracted layout
Notch with the pre-layout bias (0.2096 nA)	49.96 Hz	49.36 Hz (−0.6 Hz)
Bias for 50 Hz	0.2096 nA	0.2124 nA (+1.3%)
Bias for 60 Hz	0.2535 nA	0.2569 nA
Notch depth 50 / 60 Hz	43.1 / 41.6 dB	41.7 / 40.6 dB
Gain at 10 / 100 / 150 Hz	-7.06 / -12.70 / -14.09 dB	-7.06 / -12.99 / -14.45 dB
2 mV step: final value, overshoot	0.880 mV, 16.8%	0.887 mV, 15.8%
50 Hz rejection in the 10 + 50 Hz mix	42.3 dB	40.7 dB
THD at 5 / 10 / 20 mV	0.18 / 0.33 / 0.66 %	0.16 / 0.32 / 0.65 %
Monte Carlo usable runs	93 / 100	94 / 100
Monte Carlo f_0 (usable runs)	49.37 ± 1.78 Hz	49.77 ± 1.95 Hz
Monte Carlo depth (usable runs)	42.2 ± 1.4 dB	41.1 ± 1.5 dB
f_0 within 50 ± 5 Hz	93%	96%

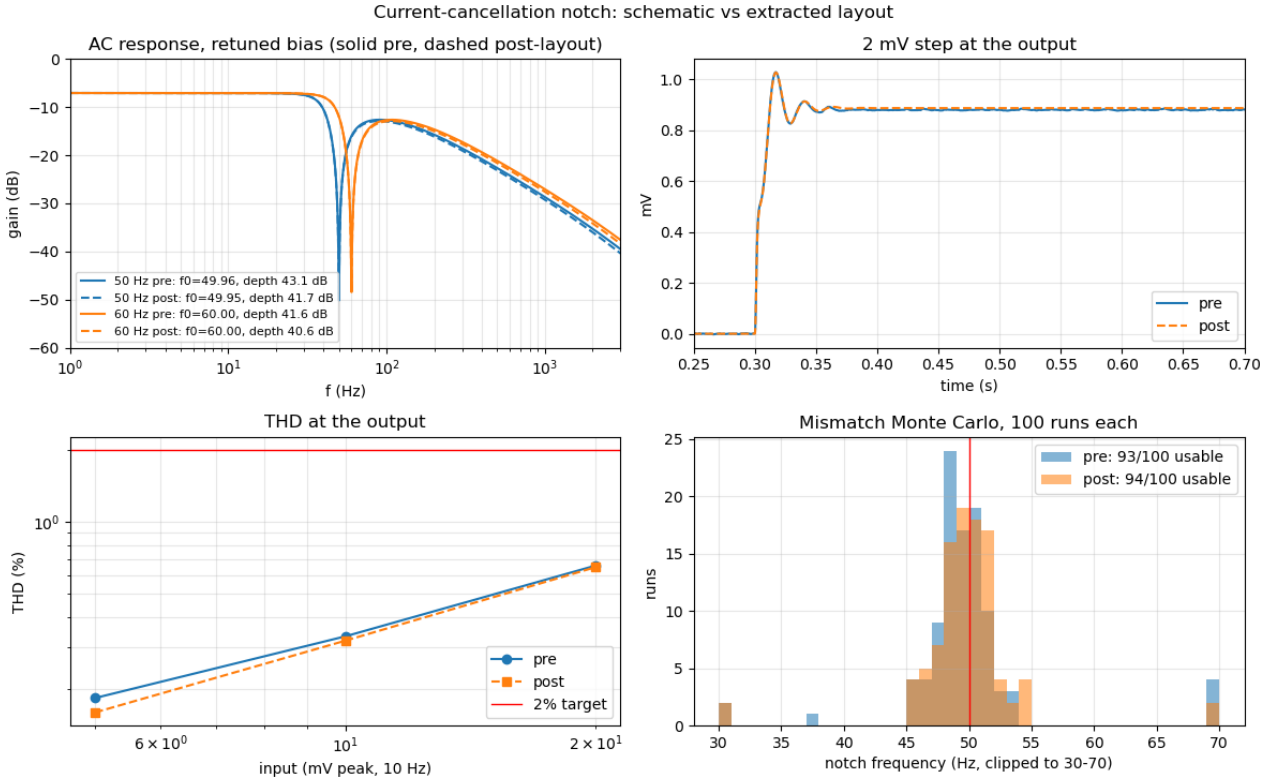


Figure 12. Schematic (solid) versus extracted layout (dashed): AC response, step, THD and Monte Carlo.

Reading the comparison. Layout parasitics move the notch down by 0.6 Hz, and a 1.3% bias increase moves it back, which is the tuning mechanism working as intended. Depth drops by about 1 dB and the band above the notch by about 0.3 dB; everything else is within simulation noise. The two Monte Carlo runs are not paired sample by sample, because the extracted netlist contains eleven transistors rather than seven, so 93 versus 94 usable runs should be read as the same yield.

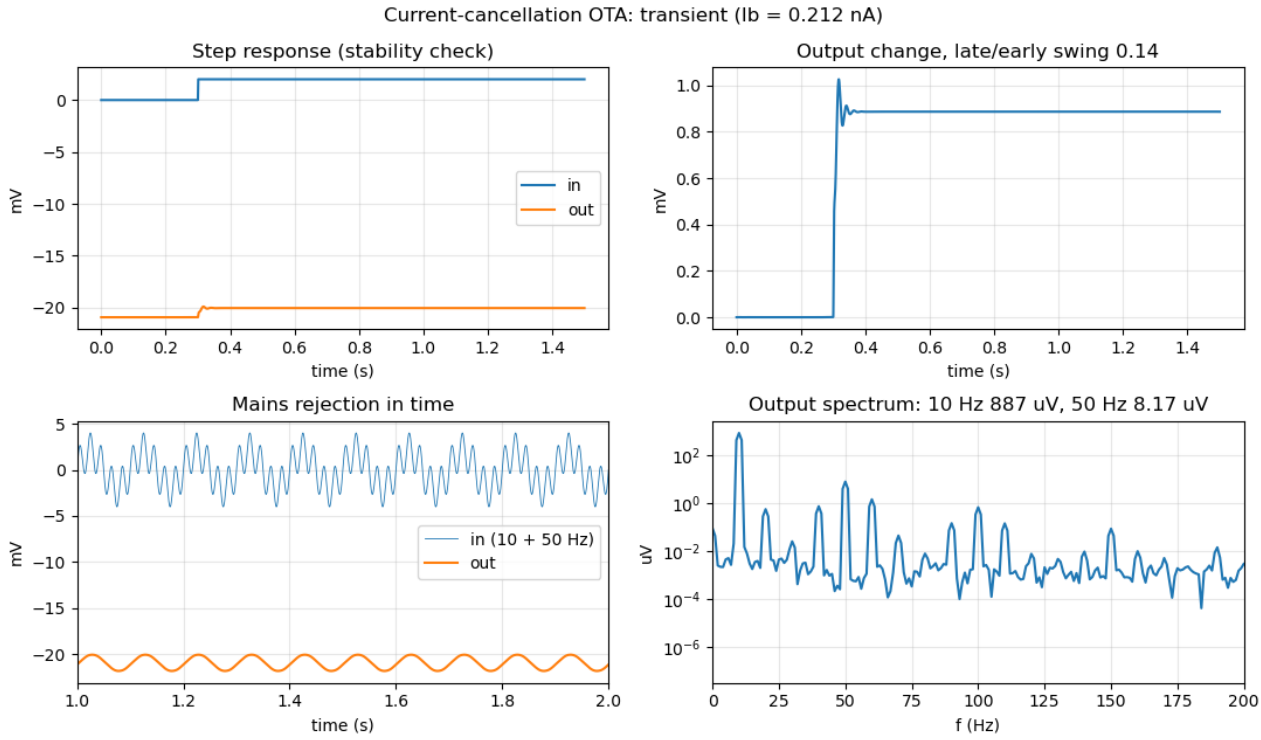


Figure 13. Post-layout transient: step and 10 + 50 Hz mix.

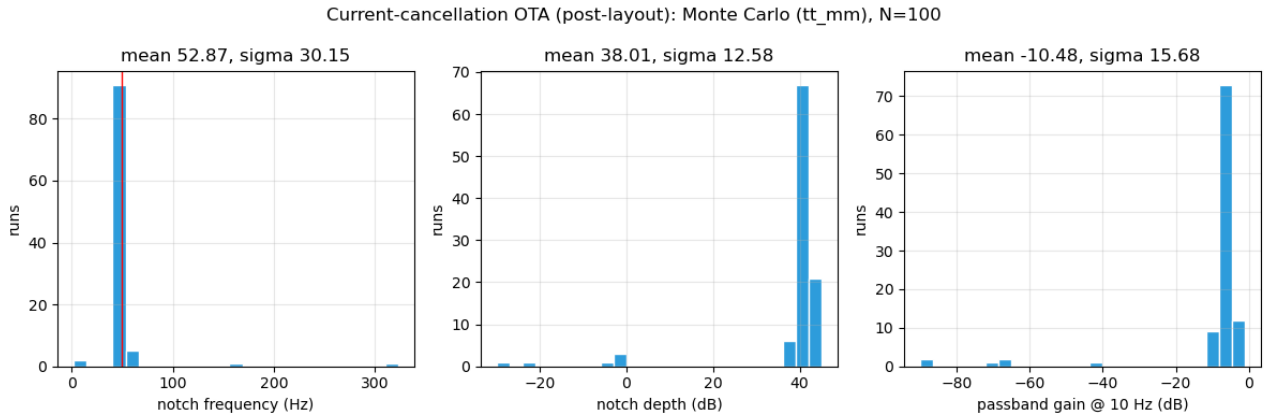


Figure 14. Post-layout Monte Carlo histograms (100 runs, mismatch).

12. Limitations and what comes next

• **Yield:** 7 of 100 mismatch runs failed. A DC-stabilising loop or offset trim would be the next design step if this style is chosen.

• **Passband above the notch:** it does not return to the low-frequency gain.

• **Process Monte Carlo:** the PDK's `mc` section failed with a model-bin error, so process variation is shown with corners, and not statistically.

• **Bias:** the testbench uses an ideal current reference. Real on-chip generation of 0.2 nA, and its spread, is not included.

• **Noise:** OTA input noise of about 150 µV rms and buffer noise of about 200 µV rms matter for small biosignals.

• **Models:** Sky130 model accuracy at picoamp device currents was not independently checked.

• **Layout scope:** only the OTA is laid out. The buffer, the MiM capacitors and the wiring between the five OTAs are still schematic-level, and wire resistance was not extracted. The input array is

interleaved but not a true common centroid and has no dummies.